ECA1803 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO4 – INVESTIGATIVE ASSIGNMENT REPORT

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Report workflo

This report follows the required workflow: REQUIREMENT → LITERATURE → ALTERNATIVES → ANALYSIS → SELECTION → SYSTEM DESIGN → VERIFICATION → FINAL DECISION. Each stage of the assignment is answered in full, and every Design Decision is completed. The table below shows where each criterion of the Scheme of Evaluation is answered.

The issued assignment leaves several numerical inputs open. The following values are adopted throughout and are stated here so that every calculation can be traced and re-run.

- **Fibre:** standard SMF-28, attenuation $\alpha = 0.2$ dB/km, dispersion $D = 17$ ps/nm/km at 1550 nm ($\beta_2 = -21.7$ ps²/km), nonlinear coefficient $\gamma = 1.3$ W^{-1}km^{-1} (effective area about 80 µm²), PMD coefficient 0.1 ps/√km.
- **Link:** unidirectional, 10 spans × 80 km = 800 km. Total span loss between amplifiers is 20 dB (16 dB fibre + 4 dB for connectors, splices, isolators, gain-flattening filter, pump coupler and ageing margin).

- **Stage 3 analytical problem:** the numeric input and output power values are not legible in the supplied copy of the brief. The values −20 dBm per channel (input) and 0 dBm per channel (output) are used. The formulas are given in general form and a sensitivity table is included so other values can be substituted directly.
- **FEC:** a 20% overhead soft-decision FEC with a pre-FEC BER threshold of 2×10^{-2} (Q = 6.25 dB) and post-FEC BER below 10^{-15}.
- **Simulation:** the figures in this report were produced with a numerical model (Gaussian-noise nonlinearity model, Monte-Carlo constellation and eye simulation) written in Python. Section 6.3 lists the OptiSystem / OptSim block settings needed to reproduce each result in a commercial simulator.

STAGE 1 – INTERPRET THE COMMUNICATION REQUIREMENT

1.1 Literature review and standards benchmarking

The standards and papers below define the technical baseline. Each row records what was taken from the source and how it influences this design.

Source	Relevant content	Influence on this design
ITU-T G.694.1 [1]	Fixed DWDM frequency grid: 12.5 – 100 GHz spacing anchored at 193.1 THz	50 GHz grid, 32 channels from 192.30 to 193.85 THz
ITU-T G.698.2 [2]	Amplified multichannel DWDM applications with single-channel optical interfaces; defines OSNR, power and dispersion parameters at reference points	Used for the link-budget structure: transmitter output, per-span power, receiver OSNR and tolerance specifications
IEEE 802.3ct [3]	100 Gb/s over DWDM systems, coherent, up to 80 km	Establishes coherent DWDM 80 km span as an accepted Ethernet reach; supports the 80 km span choice
IEEE 802.3ck [4] and OIF-CEI-112G [5]	100/200/400 Gb/s electrical interfaces with 106.25 GBd PAM4 lanes (for example 200GAUI-2)	Host-side electrical interface of the transceiver: 2 × 106.25 GBd PAM4 lanes carry the 200 Gb/s client into the DSP
OIF 400ZR IA [6]	Coherent DP-16QAM at about 60 GBd, 75 GHz spacing, 80 km, soft-decision FEC	Closest standardised precedent for DP-16QAM + hybrid DSP; this design halves the baud rate for 200 Gb/s
Kikuchi, JLT 2016 [7]; Savory, JSTQE 2010 [8]	Coherent detection theory and DSP algorithm chain (CD, MIMO, carrier recovery)	Basis for the receiver pipeline in Stage 4
Poggiolini et al., JLT 2014 [9]; Essiambre et al., JLT 2010 [10]	GN model of nonlinear interference; capacity limits of fibre	Used for the Q-factor versus launch power and reach calculations

Source	Relevant content	Influence on this design
Bromage, JLT 2004 [13]; Desurvire [12]	Raman amplification theory and design; EDFA noise and gain	Basis for the hybrid amplifier and noise-figure calculations
Reed et al., Nat. Photon. 2010 [14]; Wang et al., Nature 2018 [15]; Koos et al., JLT 2016 [16]	Silicon, thin-film lithium niobate and silicon-organic hybrid modulators	Benchmark for Vπ, bandwidth and loss figures in Stage 2
Ip et al., JLT 2008 [17]; Pfau et al., JLT 2009 [18]; Faruk and Savory, JLT 2017 [19]	Digital back-propagation; feed-forward carrier phase recovery; DSP for multilevel formats	Basis for the DSP pipeline and for rejecting full-field DBP

Literature conclusion. Published systems and standards converge on coherent detection with polarization-multiplexed QAM for DWDM transport beyond a few tens of kilometres. Direct-detection formats (NRZ-OOK, PAM4) are standardised only for short reach (up to about 10 km, mainly in the O-band), where chromatic dispersion is small. For metro/regional C-band DWDM at 200 Gb/s per wavelength, coherent DP-QAM with digital compensation is the benchmark.

1.2 Aggregate capacity

$$C_total = N_ch \times R_ch = 32 \times 200\ Gb/s = 6.4\ Tb/s\ (net\ client\ rate)$$

With a 20% FEC overhead the line rate per channel is 200 × 1.20 = 240 Gb/s, giving a gross line capacity of 32 × 240 Gb/s = 7.68 Tb/s. The 32 channels at 50 GHz occupy 32 × 50 GHz = 1.6 THz, which is about 36% of the roughly 4.4 THz C-band (1530 – 1565 nm). The design therefore leaves room to grow to 88 channels (17.6 Tb/s) in the C-band.

1.3 Key physical-layer performance metrics

Metric	Baseline value / criterion	How obtained
Net spectral efficiency	4.0 b/s/Hz (200 Gb/s / 50 GHz); 4.8 b/s/Hz gross	Rate ÷ channel spacing
Modulation (symbol) bandwidth	Symbol rate 30 GBd; occupied bandwidth 33 GHz (RRC roll-off 0.1)	Section 2.2

Metric	Baseline value / criterion	How obtained
Required SNR / OSNR (theory)	SNR 12.7 dB; OSNR 16.5 dB (0.1 nm) at pre-FEC BER 2×10^{-2}	Section 2.1 and Eq. (2)
BER / Q-factor	Pre-FEC BER $\leq 2 \times 10^{-2}$ ($Q \geq 6.25$ dB); design $Q \geq 7.7$ dB (1.5 dB margin)	$Q(\text{dB}) = 20 \log_{10}[\sqrt{2}\, \text{erfc}^{-1}(2 \cdot \text{BER})]$
Launch power per channel	0 dBm (optimum −1.2 dBm); 15 dBm total for 32 channels	Section 3.5
Required electrical modulation bandwidth	$\geq 0.7 \times 30$ GHz = 21 GHz (3 dB); target 30 GHz or more	Section 2.3
Insertion loss budget	Transmitter PIC about 19 dB; AWG mux 4.5 dB; span 20 dB; AWG demux 4.5 dB	Section 5.1
Thermal / footprint	Pluggable CFP2-class coherent module; Si PIC die of the order of 5 × 10 mm; module power about 8 W per 200 G	Section 5.2 (estimates)
CD / PMD tolerance	CD ≥ 13,600 ps/nm (800 km); mean DGD 2.8 ps, tolerate ≥ 8.5 ps	Section 4.1

1.4 Critical impairments matched to DSP functions

Impairment	Physical origin	Compensation function
Chromatic dispersion	Wavelength-dependent group velocity, D = 17 ps/nm/km	Static frequency-domain CD equalizer (FDE)
PMD and polarization rotation	Random fibre birefringence, DGD ∝ √L	Adaptive 2×2 MIMO butterfly equalizer (CMA/RDE then LMS)
Polarization-dependent loss	Isolators, couplers, WSS, amplifiers	Partly corrected by MIMO; limited by design (PDL specified at 0.2 dB or less per element)
Laser phase noise	Finite linewidth of transmit laser and LO	Carrier phase estimation (blind phase search)
Laser frequency offset	Tx/LO frequency difference up to ± 2.5 GHz	Frequency offset estimation (4th-power FFT method)

Impairment	Physical origin	Compensation function
SPM, XPM	Kerr effect, intensity-dependent refractive index	Launch power optimisation; optional digital back-propagation (DBP)
FWM	Kerr effect, phase-matched mixing products	Negligible on SMF-28 (high D); controlled by channel spacing and power
ASE noise	Amplifier spontaneous emission	Low-noise amplification (hybrid Raman + EDFA); SD-FEC
AWG / MMI crosstalk	Imperfect channel isolation	Component specification: adjacent-channel isolation of at least 25 dB
IQ imbalance, skew; limited converter bandwidth	Imperfect receiver and transmitter hardware	IQ orthogonalisation, skew correction, Tx pre-compensation

1.5 Practical trade-offs: power, integration and reliability

Platform	Strengths	Weaknesses	Verdict
Silicon photonics (SiPh)	CMOS foundry, high yield, co-integration of modulator, Ge photodiode, AWG, MMI, polarization splitter/rotator; low cost	Carrier-depletion modulator: higher loss (about 4–6 dB), Vπ about 5 V, bandwidth about 30–45 GHz, nonlinear phase response	Selected: sufficient for 30 GBd, lowest cost and best integration
Thin-film lithium niobate (TFLN)	Pockels effect, 100 GHz class bandwidth, low chirp, low Vπ (2–3 V), low loss	Less mature wafer-scale integration; no native detector or laser; higher cost	Strong option for future 90 GBd upgrade; not needed at 30 GBd
EO polymer / silicon-organic hybrid	Very low VπL (below 1 V·cm), over 100 GHz bandwidth	Thermal and photo-stability, long-term reliability, qualification risk	Not selected for a carrier-grade deployment

1.6 Selection of the performance baseline (summary)

Combining the sections above: the link must carry 6.4 Tb/s net over at least 800 km of SMF-28 with a 50 GHz grid, using a modulation format that reaches 200 Gb/s per wavelength at a symbol rate achievable with silicon photonic modulators (below about 40 GBd) and an OSNR that a hybrid amplifier chain can supply over many spans.

1.7 Investigation Requirement Statement

The DWDM system shall transport 32 wavelengths on the ITU-T G.694.1 50 GHz grid, each carrying 200 Gb/s of client data (6.4 Tb/s aggregate; 7.68 Tb/s line rate with 20% soft-decision FEC) over at least 800 km of SMF-28 in 80 km spans. It shall achieve a net spectral efficiency of at least 4 b/s/Hz and a pre-FEC BER below 2×10^{-2} (Q above 6.25 dB, SNR above 12.7 dB) at end of life with a minimum margin of 1.5 dB. Per-channel launch power shall be about 0 dBm with total launch power not above 15 dBm. The transmitter shall use a photonic integrated modulator with at least 21 GHz electrical bandwidth and drive voltage within a CMOS/SiGe driver swing. The system shall tolerate accumulated chromatic dispersion of at least 13,600 ps/nm and PMD (differential group delay) of at least 8.5 ps through digital signal processing, shall keep module power near 40 pJ/bit, and shall align with ITU-T G.694.1/G.698.2, IEEE 802.3ck (host interface) and OIF-CEI / OIF 400ZR practice.

STAGE 2 – INVESTIGATE AND SELECT THE MODULATION / MODULATOR CONFIGURATION

2.1 Comparison of candidate modulation formats

All formats are compared at the same service rate of 200 Gb/s net. The line rate includes FEC overhead (7% to 20%, depending on the format's FEC threshold). The coherent formats use the theoretical Gray-coded BER expressions below, with SNR defined per polarization and OSNR referred to a 0.1 nm (12.5 GHz) noise bandwidth:

$$BER \approx (4/\log_2 M)(1 - 1/\sqrt{M})\, Q(\sqrt{(3 \cdot SNR/(M - 1))}) \quad (1)$$

$$OSNR = SNR \times R_s / B_{ref}, \quad B_{ref} = 12.5\ GHz \quad (2)$$

$$Q(dB) = 20 \log_{10} [\, \sqrt{2} \cdot erfc^{-1}(2 \cdot BER)\,] \quad (3)$$

For BER = 2×10^{-2} these equations give Q = 6.25 dB, and the required SNR values are 6.25 dB (QPSK), 12.71 dB (16-QAM) and 18.43 dB (64-QAM). Direct-detection and differential formats are given as indicative values scaled from published 10 Gb/s receiver sensitivities (about +10 $\log_{10}$ of the baud ratio), with ±3 dB uncertainty.

Format	**Bits per symbol**	**Baud rate for 200G net**	**Channel spacing needed**	**Net SE (b/s/Hz)**	**Pre-FEC threshold**	**Required OSNR (0.1 nm)**	**Receiver / tolerance to CD**
NRZ-OOK	1	about 214 GBd (7% FEC)	≥ 400 GHz	0.5	3.8×10^{-3}	about 25 dB (indicative)	Direct detection; CD-limited to below 1 km
PAM4	2	106.25 GBd (KP4 FEC)	150 GHz	1.33	2.4×10^{-4}	about 29 dB (indicative)	Direct detection; CD limit about 650 m; needs strong driver linearity
DPSK	1	about 214 GBd	≥ 400 GHz	0.5	3.8×10^{-3}	about 22 dB (indicative)	Delay-line interferometer; CD-limited

Format	Bits per symbol	Baud rate for 200G net	Channel spacing needed	Net SE (b/s/Hz)	Pre-FEC threshold	Required OSNR (0.1 nm)	Receiver / tolerance to CD
DQPSK	2	about 107 GBd	150 GHz	1.33	3.8×10^{-3}	about 21 dB (indicative)	Two interferometers; CD-limited
16-QAM (single polarization)	4	60 GBd	75 GHz	2.67	2×10^{-2}	16.5 dB (calculated)	Coherent; CD by DSP
DP-QPSK	4	60 GBd	75 GHz	2.67	2×10^{-2}	13.1 dB (calculated)	Coherent; CD, PMD by DSP
DP-16QAM	8	30 GBd	50 GHz	4.0	2×10^{-2}	16.5 dB (calculated)	Coherent; CD, PMD by DSP
DP-64QAM	12	20 GBd	37.5 GHz	5.33	2×10^{-2}	20.5 dB (calculated)	Coherent; needs transceiver SNR above 21 dB

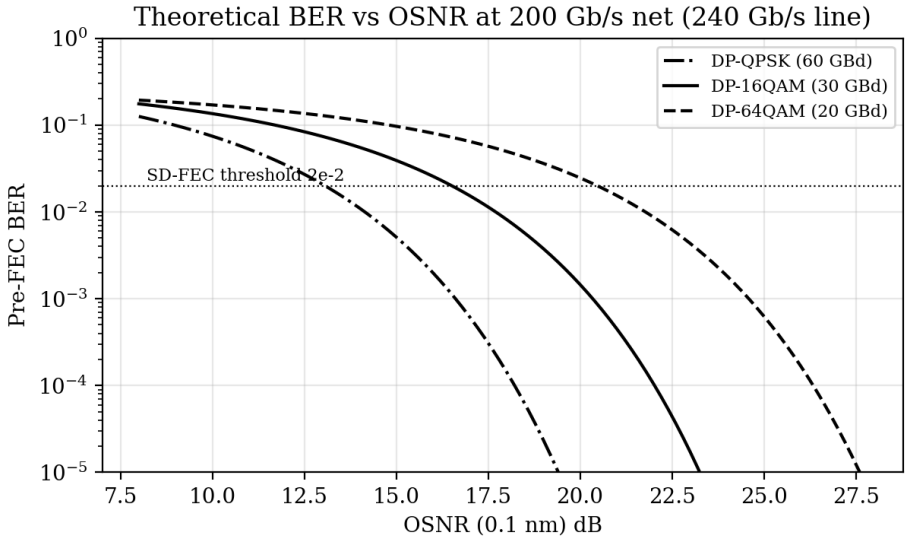


Fig. 2.1 Theoretical pre-FEC BER versus OSNR for the coherent candidates at 200 Gb/s net (240 Gb/s line rate).

Observations. (i) NRZ-OOK, PAM4, DPSK and DQPSK need symbol rates of 107 – 214 GBd and channel spacings of 150 – 400 GHz; 32 channels at these spacings would need 4.8 – 12.8 THz, which

exceeds the C-band. They also have little or no CD tolerance. (ii) DP-QPSK has the best sensitivity but needs 60 GBd and 75 GHz spacing, giving only 2.67 b/s/Hz. (iii) DP-64QAM saves bandwidth but needs about 4 dB more OSNR than DP-16QAM and a transceiver SNR above 21 dB, beyond what 5 – 6 bit effective-resolution converters and silicon modulators deliver. (iv) DP-16QAM is the balance point: 4 b/s/Hz at only 30 GBd, with an OSNR requirement (16.5 dB) that a hybrid amplifier chain supplies with margin.

2.2 Baud rate and channel spacing: DP-16QAM versus PAM4

$$R_s = R_b (1 + OH) / (m \times N_{pol}) \quad (4)$$

DP-16QAM

- Line rate: $R_b = 200 \times 1.20 = 240$ Gb/s. Bits per symbol: $m = \log_2 16 = 4$ per polarization, $N_{pol} = 2$, so 8 bits per dual-polarization symbol.
- Baud rate: $R_s = 240 / 8 =$ **30 GBd**.
- Occupied bandwidth with root-raised-cosine roll-off $\beta = 0.1$: $B = R_s(1 + \beta) = 33$ GHz. Allowing guard band for laser drift (± 2.5 GHz), AWG passband narrowing and adjacent-channel crosstalk, the channel spacing is **50 GHz** (G.694.1 grid). Net SE = 200/50 = 4 b/s/Hz.

PAM4 (direct detection)

- Line rate: $R_b = 200 \times 1.0625 = 212.5$ Gb/s (KP4-type FEC and framing). $m = 2$, $N_{pol} = 1$.
- Baud rate: $R_s = 212.5 / 2 =$ **106.25 GBd**.
- Double-sideband occupied bandwidth about $R_s(1 + \beta) \approx 117$ GHz, so the spacing is at least 125 GHz and in practice **150 GHz**. Net SE = 200/150 = 1.33 b/s/Hz, which is three times lower than DP-16QAM. Thirty-two channels would need 32 × 150 GHz = 4.8 THz, more than the C-band.
- Dispersion limit without DSP: $L_{max} \approx c / (D \lambda^2 B^2) = 3\times10^8 / (17\times10^{-6} \times (1550\times10^{-9})^2 \times (106.25\times10^9)^2) \approx 650$ m. PAM4 at 1550 nm is therefore unusable over 80 km spans without optical dispersion compensation.

Conclusion of the baud-rate calculation: DP-16QAM needs 3.5 times fewer baud (30 versus 106.25 GBd), which relaxes DAC/ADC, driver and modulator bandwidth requirements and makes dense 50 GHz DWDM possible.

2.3 Modulation mechanisms

For an interferometric modulator with differential drive voltage V across the arms, the output field is $E_{out} = E_{in} \cdot \cos(\pi V / 2V\pi)$, so the intensity transfer is $T = \cos^2(\pi V/2V\pi)$. For the linear (Pockels) effect, the half-wave voltage–length product is

$$V\pi \cdot L = \lambda\, g / (2\, n^3\, r\, \Gamma) \quad (5)$$

where g is the electrode gap, n the refractive index, r the electro-optic coefficient and Γ the optical–electrical overlap. For carrier-depletion silicon the phase shift is a plasma-dispersion effect and Vπ·L is typically 1.5 – 2.5 V·cm.

Mechanism	**Physical effect**	**Speed limit / response time**	**Vπ (typical)**	**Chirp**	**Use in this link**
Electro-optic (Pockels, $LiNbO_3$/TFLN)	Index change proportional to field	Intrinsic fs; bandwidth over 100 GHz (electrode limited)	Vπ·L about 2–3 V·cm (Vπ about 2–3 V at 1 cm)	Near zero (push-pull)	Candidate for 90 GBd upgrade
Electro-optic (plasma dispersion, Si)	Carrier depletion changes n and α	About 30–50 GHz (RC, transit time)	Vπ·L about 1.5–2.5 V·cm (about 5 V at 4 mm)	Small, residual amplitude-phase coupling	Selected
EO polymer / SOH	Large r_{33} in organic cladding on silicon	Over 100 GHz	Vπ·L below 1 V·cm	Low	Reliability risk
Acousto-optic	Photoelastic grating from acoustic wave	Acoustic transit time; below 1 GHz	Drive in RF power (not volts)	Frequency shift	Not usable at 30 GBd; used for tunable filters
Thermo-optic	$dn/dT = 1.86 \times 10^{-4}\ K^{-1}$ in silicon	Microseconds to milliseconds	Power-based, about 20 – 30 mW per π	None	Bias and phase trimming only

Mechanism	Physical effect	Speed limit / response time	Vπ (typical)	Chirp	Use in this link
Magneto-optic (Faraday)	Polarization rotation $\theta = V B L$	Slow, non-reciprocal	Not applicable	None	Isolators, rotators
Kerr (χ^3)	Quadratic EO effect, index $\propto E^2$	Fs response, but very high field	Impractical (kV class)	None	Appears as SPM/XPM impairment

Required bandwidth. For a 30 GBd Nyquist-type signal the 3 dB electrical bandwidth of the modulator and driver chain should be at least $0.7 \times R_s \approx 21$ GHz. Carrier-depletion silicon travelling-wave MZMs reach 30 – 45 GHz, and the remaining roll-off is corrected by transmitter-side digital pre-emphasis. Acousto- and thermo-optic effects are too slow for data modulation; magneto-optic effects are used only for non-reciprocal components; the Kerr effect is an impairment rather than a modulation tool at the voltages available.

Drive voltage example. For a 4 mm silicon MZM with $V\pi \cdot L = 2.0$ V·cm: $V\pi = 2.0 / 0.4 = 5$ V. For 16-QAM, each of the I and Q arms carries a 4-level signal whose swing should stay within about ± 0.6 Vπ to remain in the usable part of the cosine transfer function. A driver swing near 3 Vpp from a SiGe driver is therefore sufficient with push-pull operation.

2.4 Modulator architectures: MZI versus micro-ring resonator

A micro-ring modulator is limited by the photon lifetime of the cavity: the optical bandwidth is $\Delta f = f_0/Q$, so with a loaded Q of 8,000 the bandwidth is 193.1 THz / 8,000 = 24 GHz. Increasing bandwidth lowers Q and increases insertion loss. Silicon has a thermo-optic shift of about 0.08 nm/K for the resonance (about 10 GHz/K), so each ring must be thermally locked.

Parameter	Si MZI (travelling-wave, carrier depletion)	Si micro-ring (MRR)	EO polymer (MZI/SOH)
Electrical bandwidth	30 – 45 GHz	20 – 50 GHz (Q-limited)	over 100 GHz
Footprint	about 4 mm × 0.5 mm per MZM (about 2 mm² per DP-IQ chip arm set)	10 – 20 µm radius (about 10^{-3} mm²)	millimetre scale (MZI), micrometre scale (slot)

Parameter	Si MZI (travelling-wave, carrier depletion)	Si micro-ring (MRR)	EO polymer (MZI/SOH)
Thermal sensitivity	Low: broadband, only bias control	High: 0.08 nm/K, needs locking per ring	Moderate; polymer glass-transition limits
Insertion loss	About 4–6 dB (modulator) + coupling	1 – 3 dB on resonance	Low in slot, high coupling loss
Optical bandwidth	Over 30 nm, tolerant to laser drift	Below 0.5 nm, laser must track resonance	Broadband
Linearity for 16-QAM IQ	Cosine transfer; works with proper drive and DSP	Lorentzian, nonlinear phase, chirp; poor for QAM	Linear field-to-phase
Verdict	Selected: IQ-capable	Suited to OOK/PAM4 and dense short reach	Future option

2.5 Dual-polarization IQ modulator structure

A DP-16QAM transmitter needs four independent MZMs: two forming an I/Q pair for the X polarization and two forming an I/Q pair for the Y polarization. Each pair is a nested MZI: the I and Q MZMs are biased at the null point and combined with a 90° phase offset. The Y output is rotated 90° by an on-chip polarization rotator and combined with the X output in a polarization beam combiner. Four 4-level electrical drive signals (from 4 DACs and drivers) generate a 16-QAM constellation on each polarization. Total electrical bandwidth required per drive lane is above 21 GHz, and each lane is driven at 30 GBd.

2.6 Selection and supporting evidence

Selection criterion	DP-16QAM + Si DP-IQ MZM	Best rival and why it loses
Baud rate	30 GBd (within Si modulator bandwidth)	PAM4: 106 GBd, beyond Si MZM bandwidth
Net spectral efficiency	4.0 b/s/Hz	DP-QPSK: 2.67 b/s/Hz
OSNR requirement	16.5 dB (theory)	DP-64QAM: 20.5 dB, cannot be reached over 800 km

Selection criterion	DP-16QAM + Si DP-IQ MZM	Best rival and why it loses
CD / PMD tolerance	By DSP, no extra loss	PAM4/OOK: need DCF, 650 m limit uncompensated
Modulator bandwidth	Needed 21 GHz; available 30–45 GHz	MRR: Q-limited and nonlinear
Footprint and integration	Monolithic Si PIC with AWG, MMI, Ge PD, PBS/PR	TFLN: hybrid integration needed
Drive voltage	Vπ about 5 V, 3 Vpp driver	TFLN lower voltage but higher cost

DESIGN DECISION 1 – SELECT THE PROPOSED MODULATION / MODULATOR CONFIGURATION

I recommend dual-polarization 16-QAM (DP-16QAM) at 30 GBd (240 Gb/s line rate including 20% soft-decision FEC), generated by a silicon-photonic dual-polarization IQ Mach-Zehnder modulator (four push-pull carrier-depletion travelling-wave MZMs of about 4 mm length, $V\pi \approx 5$ V, 3 dB bandwidth of at least 30 GHz, with an on-chip polarization splitter-rotator and combiner), because it needs only 30 GBd and a 50 GHz channel (net spectral efficiency 4 b/s/Hz), which is 3.5 times lower baud and 2.7 times higher efficiency than PAM4 or DQPSK at 200 Gb/s. Its required OSNR of 16.5 dB is 4 dB below DP-64QAM and is supported by the amplifier chain (Stage 3), its dispersion and PMD tolerance is provided digitally without loss (Stage 4), and the silicon platform offers 30 – 45 GHz bandwidth, 5 V Vπ and monolithic integration with the AWG, MMI couplers and photodetectors at the lowest cost.

STAGE 3 – DEVELOP AND EVALUATE THE OPTICAL AMPLIFICATION CONFIGURATION

3.1 Comparison of amplifier technologies

Parameter	EDFA	SOA	Lumped Raman	Distributed Raman (backward)	Hybrid EDFA + Raman
Gain bandwidth	About 35 nm (C-band, 4.4 THz); L-band with other EDF	60 – 100 nm	Set by pump wavelengths; up to 100 nm	Same, multi-pump	Over 35 nm, flattened
Saturated output power	+17 to +23 dBm	+10 to +15 dBm	+20 dBm class (pump-limited)	Not applicable (gain in fibre)	+17 to +20 dBm (EDFA stage)
Noise figure	4.5 – 6 dB	7 – 9 dB	About 4–6 dB	Effective NF of −1 to −3 dB (on-off gain 8–12 dB)	Effective NF of about 0 to 2 dB
Gain tilt / flatness	About 1.5 dB ripple without GFF	Strongly wavelength dependent	Tunable by pump power/wavelengths	Tunable by multi-pump	GFF + pump tuning
Channel crosstalk	Negligible at multi-µs gain dynamics	Severe: gain recovery about 100 ps gives XGM and FWM	Low (fast gain, but pump RIN transfer in co-pumping)	Low in backward pumping	Low
Other issues	Fixed gain band; needs pump and EDF	Polarization dependence, nonlinear distortion	High pump power (1 W or more)	Pump safety, double Rayleigh scattering	Complexity, pump monitoring

Parameter	EDFA	SOA	Lumped Raman	Distributed Raman (backward)	Hybrid EDFA + Raman
Cost	Medium	Low	High	High	Highest, but fewer amplifier sites

The SOA is rejected for 32-channel 16-QAM transmission because its fast gain dynamics (about 100 ps, comparable to the 33 ps symbol period) create cross-gain modulation and four-wave mixing between channels and a high noise figure. The EDFA is the workhorse; backward distributed Raman amplification provides the lowest effective noise figure because the gain is generated inside the fibre where the signal is still strong. The hybrid combination uses both.

3.2 Analytical problem: power and span gain

Assumed values (see assumptions page): input signal level P_in = −20 dBm per channel, target output P_out = 0 dBm per channel, N = 32 channels.

$$Gain\ per\ channel:\ G = P_{out} - P_{in} = 0 - (-20) = 20\ dB\ (a\ factor\ of\ 100)$$

$$Total\ input\ power:\ P_{in,tot} = -20 + 10\log_{10}(32) = -20 + 15.05 = -4.95\ dBm\ (0.32\ mW)$$

$$Total\ output\ power:\ P_{out,tot} = 0 + 10\log_{10}(32) = +15.05\ dBm$$

$$P_{out,tot} = 10^{(15.05/10)}\ mW = 32.0\ mW = 0.032\ W$$

Result: the amplifier must deliver a total output power of 15.05 dBm (about 32 mW, 0.032 W) and a gain of 20 dB, which equals the span loss (16 dB of fibre + 4 dB of in-line components) so that each span is exactly transparent. Standard C-band in-line EDFAs have a saturated output of +17 dBm or more, so the amplifier operates with about 2 dB of output headroom, which is needed for gain flatness and transient control.

Hybrid split of the 20 dB: backward Raman on-off gain 8 dB + EDFA gain 12 dB = 20 dB. The EDFA input is −12 dBm per channel (−20 dBm + 8 dB) and its total output is 15.05 dBm. The Raman pump power is about 1.0 – 1.2 W (multi-wavelength, around 1425 – 1485 nm) for 8 dB on-off gain at the 80 km SMF-28 span.

P_in per channel (dBm)	P_out per channel (dBm)	Gain (dB)	Total output (dBm)	Total output (mW)
−20	0	20	15.05	32.0
−25	0	25	15.05	32.0
−15	0	15	15.05	32.0
−20	+2	22	17.05	50.7
−20	+3	23	18.05	63.9

General formulas for other values: G (dB) = P_out − P_in; P_tot (dBm) = P_ch (dBm) + 10 $\log_{10}$ N; P_tot (W) = 10^(P_tot/10) × 10^{-3}.

3.3 Gain flatness across the C-band and equalization

An EDFA has an intrinsic gain shape with a peak near 1532 nm and a shoulder near 1550 – 1560 nm, giving about 3 dB peak-to-peak gain variation across 1530 – 1565 nm. In a chain of N amplifiers this ripple adds in the same direction, so after N amplifiers the channel-to-channel power difference approaches N × ΔG. With ΔG = 0.3 dB residual per amplifier after flattening, 10 amplifiers would give about 3 dB of power difference; channels with lower power have lower OSNR (by the same number of dB) and channels with higher power generate more nonlinearity. Stimulated Raman scattering adds a further tilt from short to long wavelengths of about 0.1 – 0.3 dB per span for 32 channels and 15 dBm total power.

Countermeasures used:

- **Gain-flattening filter (GFF)** inside each EDFA: a thin-film or long-period-grating filter whose loss profile is the inverse of the EDFA gain shape, reducing ripple from about 3.4 dB to below 0.7 dB peak-to-peak (Fig. 3.1).
- **Raman multi-pump tuning**: pump wavelengths and powers are adjusted so that the Raman gain shape complements the EDFA shape and cancels the SRS tilt.
- **Dynamic power equalization**: a wavelength-selective switch (WSS) with per-channel attenuation every 4 – 5 spans (and at the receiver), fed by an optical channel monitor, equalizes channel power to within 0.5 dB and restores OSNR balance.
- **Automatic gain control and fast transient suppression** in the EDFA to avoid power excursions on channel add/drop.

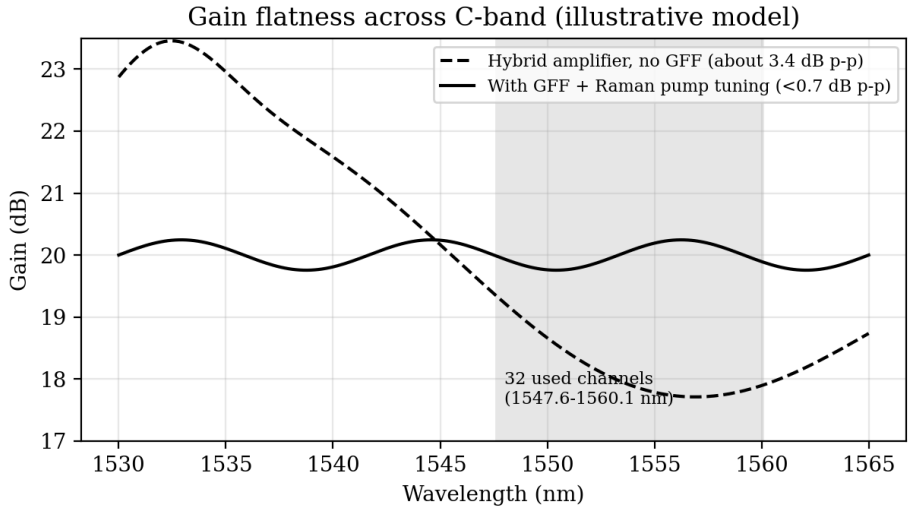


Fig. 3.1 Amplifier gain across the C-band with and without gain-flattening (illustrative model; actual values must be taken from the amplifier datasheet or simulator).

3.4 Cumulative ASE noise and OSNR degradation

The ASE power added by one amplifier of gain G and noise figure NF in reference bandwidth B_ref is P_ASE = NF · h ν · G · B_ref (with the '−1' term neglected at high gain). For N identical spans, each compensating its own loss, the OSNR at the receiver is

$$OSNR\ (dB,\ 0.1\ nm) = 58 + P_ch - L_span - NF - 10\log_{10} N \qquad (6)$$

The constant 58 dB arises from $-10\log_{10}(h\,\nu\,B_ref) = -10\log_{10}(1.28\times10^{-19}\ J \times 12.5\times10^{9}\ Hz \times 10^{3}\ mW/W) = 58$ dB at 1550 nm. With P_ch = 0 dBm and L_span = 20 dB:

Spans N	**Distance (km)**	**10 log N (dB)**	**EDFA only (NF 5.5 dB)**	**Hybrid Raman + EDFA (NF_eff 2.0 dB)**
1	80	0.0	32.5 dB	36.0 dB
5	400	7.0	25.5 dB	29.0 dB
10	800	10.0	22.5 dB	26.0 dB
14	1120	11.5	21.0 dB	24.5 dB
20	1600	13.0	19.5 dB	23.0 dB

Derivation of the hybrid noise figure. For backward Raman amplification with on-off gain 8 dB, the effective noise figure referred to the span input is about −1 dB (F_R = 0.79), followed by an EDFA with NF = 5.5 dB (F_E = 3.55) and gain G_R = 8 dB = 6.31 in front of it. By Friis' formula F_eff = F_R + (F_E − 1)/G_R = 0.79 + 2.55/6.31 = 1.20, i.e. 0.8 dB. Allowing for pump RIN transfer,

double Rayleigh scattering and splice losses, a design value of **NF_eff = 2.0 dB** is adopted, still 3.5 dB better than EDFA-only. A 3.5 dB lower noise figure corresponds to 10^0.35 = 2.2 times as many spans for the same ASE-limited OSNR.

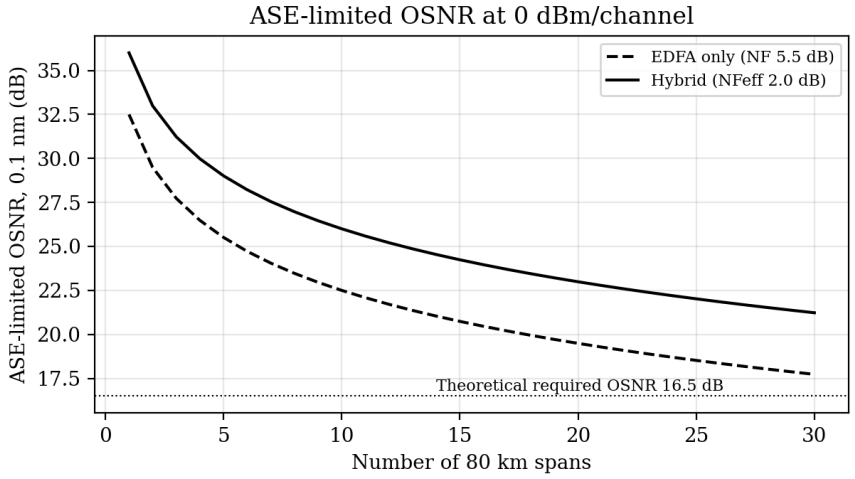


Fig. 3.2 ASE-limited OSNR versus number of 80 km spans at 0 dBm per channel.

Compared with the theoretical required OSNR of 16.5 dB, the ASE-limited OSNR of the hybrid chain at 800 km is 26.0 dB, i.e. 9.5 dB of headroom before nonlinear interference and transceiver noise are included. These additional contributions are included in the next section.

3.5 Link SNR including nonlinear interference and transceiver noise

The received SNR per symbol is obtained from the three independent noise contributions:

$$1/SNR = 1/SNR_ASE + 1/SNR_NLI + 1/SNR_TRx \quad (7)$$

SNR_ASE follows from Eq. (6) converted to the 30 GBd symbol bandwidth. SNR_NLI uses the incoherent Gaussian-noise model with coherent accumulation exponent 1.05: P_NLI = η P^3 N^1.05, where $\eta \approx 7.1\times10^3\ W^{-2}$ follows from β_2, γ, L_eff = 21.2 km, a total optical bandwidth of 1.6 THz and a flat power spectral density equal to P_ch/50 GHz. The transceiver-limited SNR_TRx is taken as 17 dB (typical for a silicon modulator, 6-bit effective-resolution converters and drivers). The optimum launch power balances ASE and nonlinearity:

$$P_opt = [\ P_ASE / (2\ \eta\ N^{1.05})\ ]^{(1/3)} \quad \text{(per-span ASE normalised)}$$

Case (10 spans, 800 km)	Optimum launch (dBm/ch)	SNR at optimum (dB)	SNR at 0 dBm/ch (dB)	BER at 0 dBm	Q at 0 dBm (dB)	Margin over 12.7 dB threshold
EDFA only (NF 5.5 dB)	−0.1	13.9	13.9	1.0×10^{-2}	7.33	1.2 dB
Hybrid Raman + EDFA (NF_eff 2.0 dB)	−1.2	14.9	14.8	5.3×10^{-3}	8.16	2.1 dB

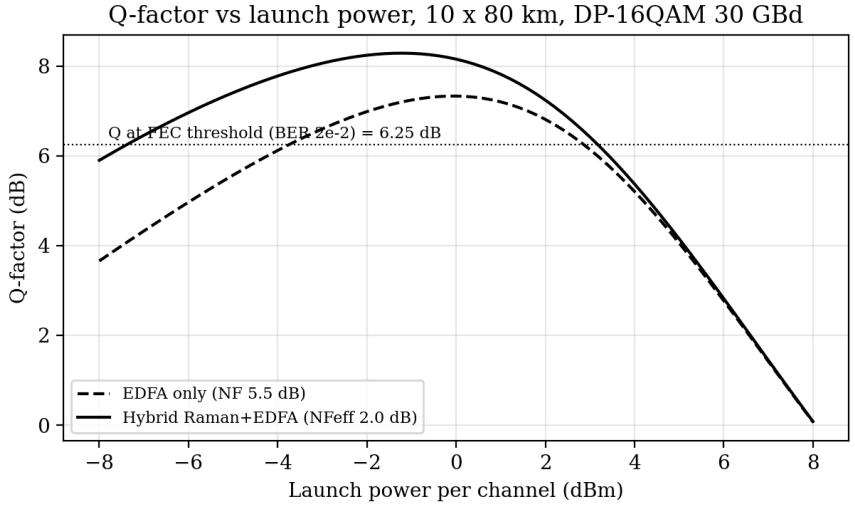


Fig. 3.3 Q-factor versus launch power per channel (10 × 80 km, DP-16QAM at 30 GBd).

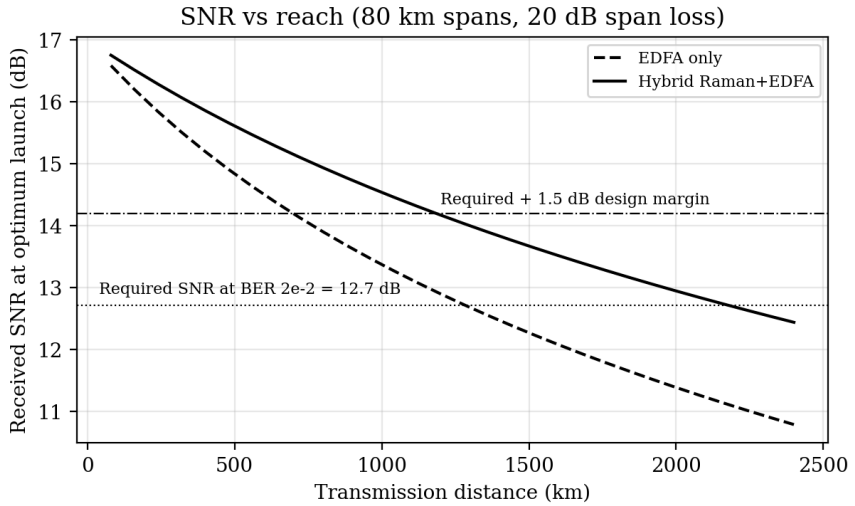


Fig. 3.4 Received SNR versus distance, with the FEC threshold and the 1.5 dB design margin.

Reach with a 1.5 dB design margin (SNR ≥ 14.2 dB): EDFA only about 8 spans (640 km); hybrid about 14 spans (1,120 km). At zero margin the limits are 16 spans (1,280 km) and 27 spans (2,160 km) respectively. The design reach of 800 km (10 spans) is therefore met with 2.1 dB margin only by the hybrid configuration, and the EDFA-only configuration misses the 1.5 dB margin criterion.

3.6 Recommended amplification architecture

- Booster EDFA after the multiplexer: output +15 dBm total (0 dBm/ch), GFF and variable optical attenuator.
- Each 80 km span: backward distributed Raman amplifier (8 dB on-off gain, multi-pump) followed by a gain-flattened EDFA (12 dB), with isolators at the input and output to suppress back-reflection and protect the pumps and Tx lasers.
- WSS-based dynamic equalizer every 5 spans and at the receiver; optical channel monitor and automatic gain control.
- Last amplifier acts as pre-amplifier ahead of the AWG demultiplexer.

DESIGN DECISION 2 – SELECT THE PROPOSED AMPLIFICATION CONFIGURATION

I recommend a hybrid backward distributed Raman plus gain-flattened EDFA amplifier in every 80 km span (8 dB Raman on-off gain + 12 dB EDFA gain = 20 dB span gain, output 15.05 dBm total / 0 dBm per channel, effective noise figure about 2 dB, GFF plus pump tuning, with a WSS equalizer every five spans), because it meets the link power budget exactly (20 dB gain for 20 dB span loss, 32 mW total output within the EDFA saturation power) and gives 3.5 dB lower effective noise figure than EDFA-only. At 800 km it reaches an SNR of 14.8 dB against the 12.7 dB threshold (2.1 dB margin, BER 5.3×10^{-3}), whereas EDFA-only leaves only 1.2 dB, below the 1.5 dB design margin. The ASE-limited OSNR of 26.0 dB comfortably exceeds the required 16.5 dB, gain ripple stays below 0.7 dB per amplifier after flattening and is controlled by WSS equalization, and the SOA is excluded because of its crosstalk and 7 – 9 dB noise figure.

STAGE 4 – INVESTIGATE AND SELECT THE WDM IMPAIRMENT / DSP COMPENSATION CONFIGURATION

4.1 Physical mechanisms and magnitudes on SMF-28

Impairment	Mechanism	Quantified for this link
Chromatic dispersion (CD)	Group-velocity dispersion $\beta_2 = -D\lambda^2/(2\pi c) = -21.7$ ps²/km. Different spectral components arrive at different times, causing inter-symbol interference.	Accumulated: 17 × 800 = 13,600 ps/nm. Signal bandwidth $\Delta\lambda = \lambda^2\Delta f/c = 0.264$ nm (33 GHz), so broadening = 13,600 × 0.264 = 3.6 ns, which is about 108 symbol periods (T = 33 ps).
Polarization mode dispersion (PMD)	Random birefringence splits the signal into two polarizations with differential group delay (DGD).	Mean DGD = 0.1 × √800 = 2.8 ps (0.085 T); maximum (about 3 × mean) = 8.5 ps.
Self-phase modulation (SPM)	Intensity-dependent phase, $\varphi_NL = \gamma P L_eff$	L_eff = 21.2 km: $\varphi_NL = 1.3 \times 1$ mW × 21.2 km = 0.028 rad per span at 1 mW per channel; accumulates over 10 spans.
Cross-phase modulation (XPM)	Phase shift on one channel from the intensity of neighbours; converted to noise through dispersion	Dominant nonlinear contribution in 32-channel 50 GHz WDM; included in the GN model with total bandwidth 1.6 THz.
Four-wave mixing (FWM)	Mixing products at f_i + f_j − f_k; efficiency depends on phase mismatch $\Delta\beta = \beta_2(2\pi\Delta f)^2$	For $\Delta f = 50$ GHz, $\Delta\beta = 2.1$ rad/km, coherence length about 1.5 km, efficiency $\eta \approx \alpha^2/(\alpha^2 + \Delta\beta^2) = 4.6 \times 10^{-4}$ (−33 dB): negligible because D is high on SMF-28.
Polarization-dependent loss (PDL)	Polarization-dependent attenuation in isolators, couplers, WSS, amplifiers	Accumulates as a random walk, about 0.2 dB × √N; cannot be removed by DSP and reduces SNR, so it is controlled by component specification (0.2 dB or less each).
ASE noise and crosstalk	Amplifier noise; AWG/MMI imperfect isolation	Addressed in Stage 3 and by AWG isolation of at least 25 dB for adjacent channels.

4.2 Optical-domain versus digital compensation

Criterion	DCF (optical)	Optical phase conjugation (OPC)	Coherent DSP (digital)
Principle	Fibre with D of about −110 ps/nm/km	Mid-link spectral inversion so the second half undoes the first	Digital all-pass filter (FDE) at the receiver
Cost for 80 km span	L_DCF = 17 × 80 / 110 = 12.4 km per span (124 km over 10 spans); about 6 dB extra loss per span at 0.5 dB/km, requiring more gain and raising NF	Needs HNLF or PPLN conversion, pumps and symmetric power map: complex and not practical for each channel	No optical loss; area/power in the DSP ASIC
Nonlinearity	Small effective area (about 20 µm²) increases nonlinearity	Can compensate Kerr effects in principle	Optional DBP
Flexibility	Fixed per span and fibre type; no PMD compensation	Static, needs identical spans; no PMD compensation	Adaptive to any fibre, handles PMD, polarization and phase noise
Latency	About 60 µs per 12 km of DCF	Low	Tens of ns to µs in ASIC
Verdict	Not selected	Not selected (laboratory technique)	Selected

4.3 Operation of the coherent DSP blocks

(a) Static chromatic dispersion equalization (FIR / frequency-domain)

The fibre transfer function is $H(\omega) = \exp(-j \cdot (D\lambda^2 z/(4\pi c))\, \omega^2)$. The receiver applies its inverse. The number of time-domain FIR taps for a sampling period T_s is $N = 2\lfloor |D|\lambda^2 L/(2cT_s^2) \rfloor + 1$. At 2 samples per symbol (T_s = 16.7 ps) and L = 800 km, $N \approx 393$ taps (about 981 taps for 2,000 km). Because direct convolution with hundreds of taps is expensive, the equalizer is implemented in the frequency domain (overlap-save FFT, block length 1,024 – 2,048), whose cost grows only logarithmically with the dispersion.

(b) Adaptive polarization demultiplexing and PMD compensation (2×2 MIMO)

A butterfly FIR structure with four adaptive filters h_xx, h_xy, h_yx, h_yy produces the outputs x_out = h_xx^H x + h_xy^H y and y_out = h_yx^H x + h_yy^H y. The constant-modulus algorithm (CMA) updates the taps with $\varepsilon = R - |y_{out}|^2$ and $h \leftarrow h + \mu\, \varepsilon\, y_{out}\, x^*$. Since 16-QAM is not a constant-modulus format, CMA (or radius-directed equalization, RDE) is used for blind convergence, after which the equalizer switches to a decision-directed LMS update that also removes residual ISI. With T/2-spaced taps, 21 taps cover ±10 samples = ±167 ps, which is far larger than the 8.5 ps worst-case DGD, leaving margin for residual CD, filter shaping and PDL.

(c) Digital back-propagation (DBP)

DBP solves the nonlinear Schrödinger equation backwards using the split-step Fourier method: in each step the signal passes through a linear dispersion operator $\exp(-j\beta_2\omega^2\Delta z/2)$ with the sign reversed, followed by a nonlinear phase rotation $\exp(+j\gamma_{eff}|A|^2\,\Delta z)$. With K steps per span and N_s spans, the number of FFT pairs is K × N_s (for example 2 × 10 = 20 steps, i.e. 40 FFT/IFFT per block), which is an order of magnitude more computation than the static CD block. Single-channel DBP compensates only SPM and the signal's own nonlinearity; XPM from the other 31 channels cannot be removed, so the gain is small (modelled 0.2 dB, literature up to about 0.5 dB). Full-field DBP of all 32 channels would require coherent detection and sampling of 1.6 THz of bandwidth and would not be implementable.

DBP option	**SNR gain (literature / this model)**	**Relative computation**	**Relative DSP power**	**Decision**
None (CD FDE only)	0 dB	1×	1× (baseline, about 8 W per 200G module)	Adopted
Single-channel DBP, 1–2 steps/span	0.2 dB (model), up to 0.5 dB (literature)	About 10 – 20×	+ 30% to + 100% DSP power	Optional future firmware/ASIC upgrade
Few-channel DBP (3 – 5 channels)	About 1 – 2 dB (literature)	Over 50×	Several times	Rejected
Full-field DBP (32 channels)	Highest	Not implementable	Not implementable	Rejected

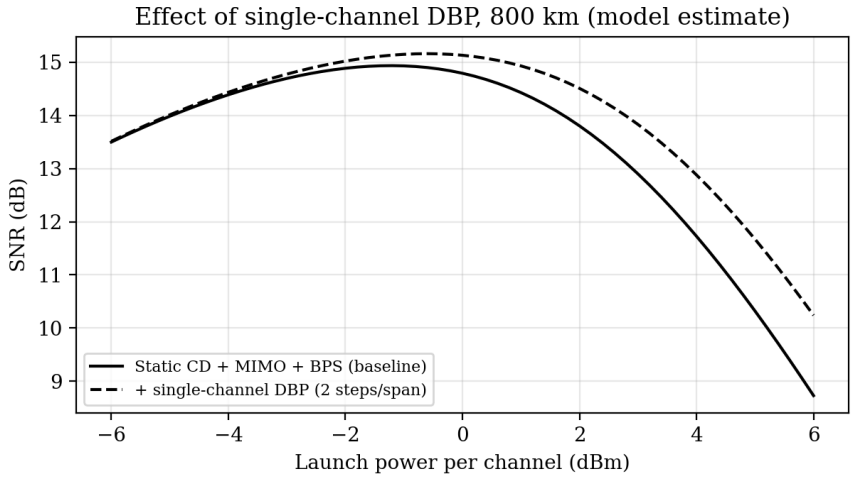


Fig. 4.1 SNR versus launch power with and without single-channel DBP at 800 km (model estimate): the gain is only about 0.2 dB.

(d) Frequency-offset and carrier-phase estimation

The offset between the transmit laser and the local oscillator (up to ± 2.5 GHz) is estimated from the spectral peak of the fourth power of the signal and removed by a digital frequency shift. The remaining laser phase noise is tracked by a feed-forward blind phase search (BPS) with 32 test phases and a 10 – 20 symbol averaging window; BPS is hardware-efficient and works for all square QAM formats. Linewidth tolerance is set by $\Delta\nu \cdot T_s \approx 10^{-4}$ for about 1 dB penalty; with a combined Tx + LO linewidth of 200 kHz at 30 GBd, $\Delta\nu \cdot T_s = 6.7 \times 10^{-6}$, well inside this limit.

4.4 Recommended coherent DSP receiver pipeline and justification of the order

#	**Block**	**Function**	**Why in this position**
1	ADC (60 GSa/s, 2 samples/symbol) + front-end correction	IQ orthogonalization (Gram-Schmidt), DC removal, IQ skew correction	Removes deterministic hardware errors first so all later algorithms see ideal I/Q signals
2	Static CD equalizer (FDE, about 393 taps equivalent)	Removes accumulated 13,600 ps/nm	CD is the largest, deterministic, polarization-independent impairment; removing it first keeps the MIMO equalizer short (21 taps) and lets timing recovery see an undistorted clock tone

#	Block	Function	Why in this position
3	Timing recovery (Gardner / Godard clock phase detector, interpolator)	Corrects sampling clock offset and jitter	Needs CD-free signal; must precede the symbol-rate MIMO output
4	2×2 MIMO adaptive equalizer: CMA/RDE for blind start, then decision-directed LMS	Polarization demultiplexing, PMD, residual ISI, PDL partial compensation	Cannot run after phase recovery because polarization and PMD compensation must be done before the signal is coherent enough for phase tracking; CMA works without phase information
5	Frequency offset estimation (4th-power FFT)	Removes Tx–LO frequency offset	Required before phase tracking, which only follows small phase variations
6	Carrier phase estimation (BPS)	Tracks laser phase noise	Needs polarization-separated, ISI-free symbols and a small residual frequency offset
7	Symbol decision, demapping, soft-decision FEC (20% overhead, threshold 2×10^{-2})	Outputs 200 Gb/s client data with post-FEC BER below 10^{-15}	Final stage; corrects the remaining random errors

If DBP is later enabled it replaces block 2 (CD equalizer) by a combined linear/nonlinear step chain and is placed before the polarization MIMO because propagation acts on the physical fields before polarization demultiplexing.

4.5 Expected improvement in Q-factor, BER and reach

Condition (800 km, hybrid amplifier chain)	Effect on signal	Result
No DSP, direct sampling of the coherent receiver	3.6 ns CD broadening (108 symbols of ISI); random polarization	BER about 0.5, no recovery possible

Condition (800 km, hybrid amplifier chain)	Effect on signal	Result
CD equalizer only	ISI removed; polarization still mixed; PMD uncompensated	No convergent decision; BER above FEC threshold
CD + MIMO + FOE + BPS (selected)	All linear impairments removed; transceiver, ASE and nonlinear noise remain	SNR 14.8 dB, BER 5.3×10^{-3}, Q = 8.16 dB (1.9 dB above the 6.25 dB threshold), error-free after SD-FEC
+ single-channel DBP (optional)	Removes part of SPM	SNR about +0.2 dB, reach about +5%

With the selected DSP chain the receiver tolerates all CD up to the full 800 km, DGD up to far beyond the 8.5 ps maximum, and linewidth sums up to hundreds of kHz. Reach with 1.5 dB margin is 1,120 km (hybrid) and the nonlinear penalty is controlled by launch-power optimisation rather than by DBP.

4.6 Impairment trade-off summary

Impairment	Compensation chosen	Residual / cost
CD	FDE, about 393 taps (800 km)	DSP area and power; negligible optical cost
PMD and polarization	2×2 MIMO CMA → LMS, 21 T/2 taps	Convergence time of microseconds; PDL partly left
Laser phase noise and frequency offset	FOE + BPS	Needs ITLA linewidth ≤ 100 kHz
SPM / XPM / FWM	Launch-power optimisation (about 0 dBm/ch); FWM naturally low	Nonlinear SNR limit; DBP rejected for cost versus 0.2 dB gain
ASE	Hybrid amplifier + SD-FEC	Overhead of 20%
Hardware imperfection	Tx pre-emphasis, IQ correction	ADC/DAC resolution sets SNR_TRx of about 17 dB

DESIGN DECISION 3 – SELECT THE DSP COMPENSATION CONFIGURATION

I recommend a fully digital coherent receiver DSP chain in the order IQ correction → static frequency-domain CD equalization (about 393 taps at 800 km, scalable to about 981 taps at 2,000 km) → timing recovery → 2×2 MIMO butterfly equalizer (CMA/RDE blind start then decision-directed LMS, 21 T/2-spaced taps) → 4th-power frequency-offset estimation → blind-phase-search carrier recovery (32 test phases) → soft-decision FEC with 20% overhead, with no optical dispersion compensation and with digital back-propagation left out of the baseline, because this chain removes the 13,600 ps/nm of dispersion and up to 8.5 ps DGD without adding any optical loss or noise (DCF would add about 6 dB loss per span) and brings the 800 km link to SNR 14.8 dB, BER 5.3×10^{-3} and Q = 8.16 dB, which is 2.1 dB above the FEC threshold. DBP adds only about 0.2 dB (up to 0.5 dB in the literature) for 10 – 20 times more computation, and the nonlinear impairment is better controlled by operating at the optimum launch power of about −1 to 0 dBm per channel.

STAGE 5 – EVALUATE THE INTEGRATED WDM TRANSMISSION STRATEGY

5.1 Integration of transmitter, photonic circuits, amplifiers and DSP receiver

The complete link (Fig. 6.1 in Stage 6) is built from 32 identical transceiver channels. Each transmitter combines an ITLA laser, a silicon DP-IQ modulator, DAC/driver electronics and transmit DSP. The 32 channels are combined in a 32:1 silicon AWG multiplexer, boosted by an EDFA, sent through 10 hybrid Raman/EDFA spans of 80 km, pre-amplified, demultiplexed by an AWG, and detected by 32 intradyne coherent receivers followed by ADCs and the DSP chain.

Photonic integrated circuit components

- **Multimode interference (MMI) couplers:** self-imaging beat length $L_\pi = 4\, n_r\, W_e^2 / (3\,\lambda_0)$. For silicon ($n_r \approx 2.85$, effective width $W_e = 6$ μm): $L_\pi \approx 88$ μm, so a 2×2 3-dB MMI is about $3L_\pi/2 \approx 130$ μm long with 0.2 – 0.5 dB excess loss. MMIs split the laser into four modulator arms (1×2 stages) and form the 90° hybrids in the receiver.
- **Directional couplers (coupled-mode theory):** cross-coupled power $P_{cross} = \sin^2(\kappa L)$, so a 3 dB coupler needs $\kappa L = \pi/4$. They are used for power taps (monitor photodiodes) and Mach-Zehnder interferometer arms.
- **Arrayed waveguide grating (AWG):** 32 channels, 50 GHz spacing, insertion loss 4.5 dB, adjacent-channel crosstalk of at most −25 dB, flat-top passband of at least 35 GHz (3 dB). The phase difference between neighbouring array waveguides, determined by coupled-mode and path-length design, focuses each wavelength at a different output port.
- **Isolators and Faraday rotators:** a 45° Faraday rotation $\theta = V\,B\,L$ between polarizers gives a non-reciprocal isolator with isolation of at least 30 dB and insertion loss of at most 0.5 dB. Isolators protect the lasers and the Raman/EDFA pumps from back-reflection and Rayleigh backscatter. Optical circulators are not placed in the main signal path; PMD compensators are not needed because the MIMO equalizer compensates PMD digitally.

Power budget and effect of coupler loss on launch power

Element	**Loss / gain**	**Per-channel level (dBm)**
ITLA laser output	+16 dBm	+16
Si DP-IQ modulator (null-bias modulation loss 6 dB, waveguide 5 dB, MMIs 2 dB, PBC/rotator 1 dB, fibre coupling 3 + 2 dB)	about −19 dB	−3

Element	Loss / gain	Per-channel level (dBm)
AWG multiplexer (32:1)	−4.5 dB	−7.5
Isolator + VOA	−0.5 dB	−8.0
Booster EDFA (gain 8 dB, total output 15.05 dBm)	+8 dB	0 (launch)
Span: 80 km SMF + components	−20 dB	−20
Raman (8 dB) + EDFA (12 dB)	+20 dB	0 (next span)
Last amplifier / pre-amplifier, WSS, AWG demux 4.5 dB	net about −2 to −7 dB	about −7 at ICR input (ICR range −15 to 0 dBm)

Effect of MMI-based multiplexing. If the 32 channels were combined by a passive MMI/star-coupler tree instead of an AWG, the combining loss would be $10 \log_{10} 32 = 15$ dB plus about 2 dB excess loss, i.e. 17 dB instead of 4.5 dB. The per-channel level at the booster input would fall from −8 dBm to −20.5 dBm. Every dB lost at this point either lowers the launch power (reducing SNR_ASE one-for-one) or has to be recovered by more booster gain at its noise figure. With a 20.5 dB booster the transmit-side OSNR would be $58 - 20.5 - 5.5 = 32.0$ dB, which lowers the 800 km ASE OSNR from 25.9 dB (AWG, transmit OSNR 44.5 dB) to 25.0 dB, a loss of about 0.9 dB, and would also need a higher-power booster stage. The AWG is therefore selected for the 32-channel multiplexer, while MMIs are used for on-chip splitting and for the coherent hybrids, where their small loss and broadband behaviour are advantageous.

5.2 Global performance parameters

Parameter	Value	Basis
Aggregate throughput	6.4 Tb/s net (7.68 Tb/s line)	32 × 200 Gb/s (× 1.2 FEC)
Net spectral efficiency	4.0 b/s/Hz (4.8 gross)	200 Gb/s / 50 GHz
System SNR at 800 km (0 dBm/ch)	14.8 dB (optimum −1.2 dBm: 14.9 dB)	Eq. (7), GN model
System Q-factor / BER (pre-FEC)	Q = 8.16 dB; BER = 5.3×10^{-3}	Eqs. (1), (3)

Parameter	Value	Basis
OSNR margin over FEC threshold	2.1 dB (SNR 14.8 versus 12.7 dB); equivalent OSNR 18.6 versus 16.5 dB	Eq. (2): OSNR = SNR + 3.8 dB
Power consumption	About 8 W per 200G coherent module (40 pJ/bit) × 32 = 256 W; amplifiers about 18 W per span site × 10 + 10 W booster = 190 W; total about 446 W, i.e. about 70 pJ/bit	Engineering estimate from 400ZR-class module power (about 15 – 20 W per 400G) and typical EDFA/Raman unit power
Footprint	32 pluggable coherent modules (CFP2-class, 41.5 × 107.5 × 12.4 mm), AWG mux and demux (1U each), booster/pre-amplifier (1 – 2U); in-line sites about 2U; Si PIC die of the order of 5 × 10 mm	Estimate

5.3 Inter-dependencies

How the modulation order sets amplifier OSNR requirement and DSP complexity. At the same 240 Gb/s line rate, the required OSNR rises from 13.1 dB (DP-QPSK) to 16.5 dB (DP-16QAM) to 20.5 dB (DP-64QAM): every doubling of bits per symbol per polarization costs 3.4 – 4 dB. Each 1 dB increase in required OSNR cuts the number of spans by 21% (10^0.1) for the same amplifier noise figure, so moving from 16QAM to 64QAM would cut the reach to about 3 spans (240 km) even with the hybrid chain, because the transceiver SNR (about 22 dB for this to work) and the nonlinear limit leave almost no margin. Higher order also tightens the phase-noise tolerance, requires ADCs/DACs with higher effective resolution (about 5.5 bits for 16QAM versus 7 or more for 64QAM), a decision-directed equalizer and a finer BPS grid, raising DSP area and power. DP-16QAM is therefore the balanced operating point: moderate OSNR, moderate DSP.

How the amplifier noise figure sets the format. The hybrid Raman/EDFA chain reduces NF from 5.5 to 2.0 dB, which lifts the 800 km OSNR by 3.5 dB. This is the difference between a 1.2 dB margin (EDFA only) and a 2.1 dB margin (hybrid) for DP-16QAM, and it is the only way to keep a 1.5 dB design margin at 800 km.

How MMI/AWG losses impact launch power: see Section 5.1 (a 12.5 dB larger combining loss lowers booster-input power from −8 to −20.5 dBm and costs about 0.9 dB OSNR).

5.4 Trade-offs: reach, channel density, dynamic range and feasibility

Option (200 Gb/s per channel)	Spacing / 32-ch band	Reach with 1.5 dB margin (model)	Channels in C-band	Feasibility
DP-QPSK, 60 GBd	75 GHz / 2.4 THz	Over 3,000 km (upper bound; ignores filtering and PDL)	About 58	Needs 60 GBd modulators and converters (beyond silicon MZM bandwidth), 2× DSP and ADC speed
DP-16QAM, 30 GBd	50 GHz / 1.6 THz	1,120 km (hybrid); 640 km (EDFA only)	About 88	Feasible with Si PIC and CMOS DSP
DP-64QAM, 20 GBd	37.5 GHz / 1.2 THz	About 240 km (3 spans)	About 117	Needs transceiver SNR above 21 dB; not feasible with Si MZM and 5-bit converters

Dynamic range. With the hybrid chain, the received SNR stays within 0.5 dB of its maximum over a launch window of about −3.8 to +0.9 dBm per channel (4.7 dB wide), so the design tolerates ± 2 dB power errors from gain ripple, ageing and add/drop transients. The ICR input range (−15 to 0 dBm per channel) covers the expected −7 dBm with ± 7 dB tolerance. Channel density versus reach: the 50 GHz grid yields 4 b/s/Hz and 17.6 Tb/s per C-band fibre, at the cost of reduced reach compared with 75 GHz DP-QPSK. Deployment feasibility: all elements (Si PIC, 7 nm DSP, EDFA, backward Raman) are commercially available.

5.5 Critical failure points, scalability limits and standards compliance

Risk / limit	Consequence	Mitigation
Raman pump failure or fibre cut near pump (eye safety)	Loss of 8 dB gain in the span; high optical power at connectors	Automatic power reduction on loss-of-signal; EDFA gain headroom; pump monitoring
Gain ripple and SRS tilt accumulate	Channel OSNR imbalance, weakest channel limits	GFF, pump tuning, WSS equalizer every 5 spans, channel monitor
Transient power changes on channel add/drop	Q penalty on surviving channels	Automatic gain control, fast transient suppression

Risk / limit	Consequence	Mitigation
Tx/LO laser drift and Si PIC thermal drift	Frequency offset, MZM bias drift	Wavelength locker, thermal control, automatic bias control; FOE tolerates ± 2.5 GHz
PDL accumulation	SNR penalty not recoverable by DSP	Component PDL of 0.2 dB or less, monitored in commissioning
Converter resolution	Limits transceiver SNR to about 17 dB (ceiling for 16QAM)	Choose ADC/DAC with effective resolution of at least 5.5 bits
Scalability: C-band full (88 channels, 17.6 Tb/s); further growth	Upgrade needs new band or higher rate	Add L-band amplifiers; upgrade to 400G (DP-16QAM at 60 GBd on 75 GHz, as in 400ZR) or TFLN modulators

Standards. Channel plan: 50 GHz grid anchored at 193.1 THz, ITU-T G.694.1; amplified multichannel DWDM link structure and OSNR/power parameters follow ITU-T G.698.2 practice. Host electrical interface of the module: IEEE 802.3ck / OIF-CEI-112G (2 × 106.25 GBd PAM4 lanes, 200GAUI-2). The line side is architecturally aligned with IEEE 802.3ct (coherent DWDM over 80 km spans) and OIF 400ZR (DP-16QAM coherent with soft-decision FEC). Note that 30 GBd DP-16QAM at 200 Gb/s is not itself a published IEEE or OIF application code; formal compliance would require conformance testing of the final module.

DESIGN DECISION 4 – CREATE THE INTEGRATED WDM STRATEGY

I recommend adopting the integrated 32-channel, 6.4 Tb/s DP-16QAM (30 GBd, 50 GHz grid) architecture with silicon-photonic DP-IQ transmitters, a 32:1 AWG multiplexer and booster, ten 80 km hybrid backward-Raman plus gain-flattened-EDFA spans at 0 dBm per channel, and a coherent receiver with CD-FDE, 2×2 MIMO, FOE, BPS and 20%-overhead SD-FEC for network deployment up to 800 km (with a demonstrated capability of up to about 1,100 km at 1.5 dB margin), because the cross-stage evidence shows a consistent, margin-positive design: net spectral efficiency 4 b/s/Hz; 2.1 dB SNR margin at 800 km (SNR 14.8 dB versus the 12.7 dB threshold, BER 5.3×10^{-3}); amplifier chain with 20 dB gain, 32 mW total output and 26 dB ASE-limited OSNR; digital compensation of 13,600 ps/nm and 8.5 ps DGD without optical loss; module power of about 40 pJ/bit (about 70 pJ/bit with amplifiers); and a monolithic PIC with AWG and MMI couplers, using established 400ZR-class technology.

STAGE 6 – VERIFY AND CRITICALLY EVALUATE THE COMPLETE WDM SYSTEM

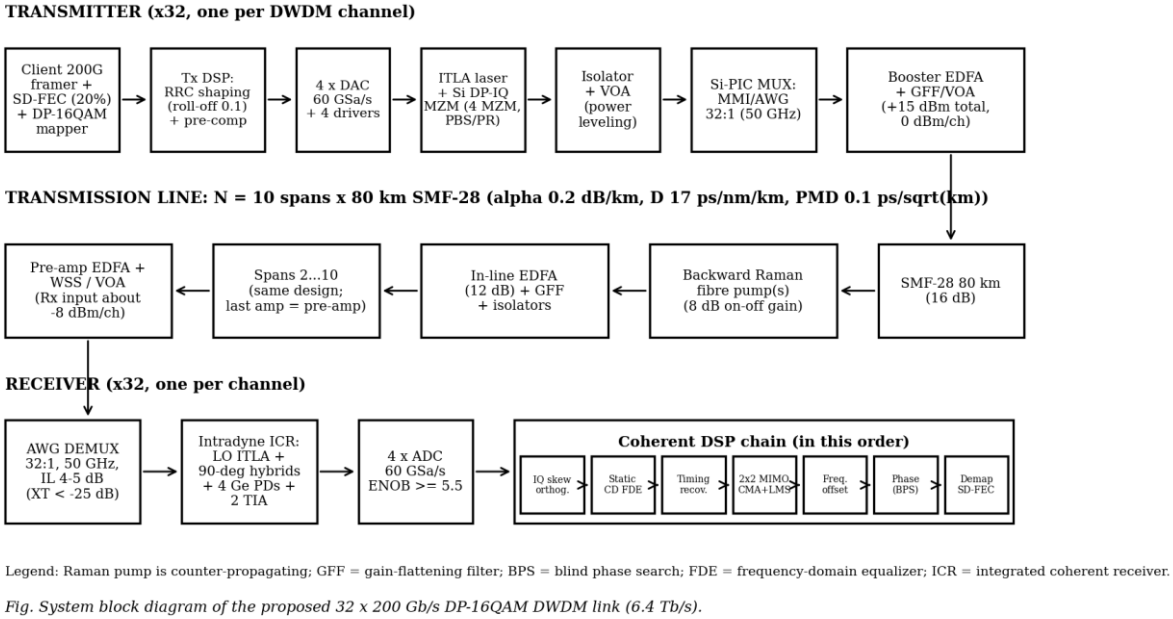
6.1 Final verification table: target versus achieved

Requirement / performance parameter	**Specified / target value**	**Achieved / evidence value**	**Status**
Aggregate WDM capacity	6.4 Tb/s (32 ch × 200 Gb/s)	32 × 200 Gb/s = 6.4 Tb/s net; 7.68 Tb/s line rate with 20% FEC	Met
Spectral efficiency and modulation choice	DP-16QAM; at least 4 b/s/Hz	DP-16QAM, 30 GBd, 8 bits/symbol, 50 GHz grid; net SE 4.0 b/s/Hz (4.8 gross); occupied bandwidth 33 GHz	Met
Modulator bandwidth	At least 21 GHz (0.7 × R_s)	Si travelling-wave MZM, 3 dB bandwidth of 30 – 45 GHz (30 GHz design value)	Met
Modulator drive and footprint	Compact PIC footprint	$V\pi \approx 5$ V (4 mm, $V\pi L = 2$ V·cm), 3 Vpp driver; Si PIC die of the order of 5 × 10 mm including AWG/MMI/Ge PD	Met
Amplifier gain	20 dB to match 20 dB span loss	Raman 8 dB + EDFA 12 dB = 20 dB; total output 15.05 dBm (32 mW)	Met
Gain flatness	Channel power spread below 1 dB	Below 0.7 dB p-p per amplifier with GFF; at most 0.5 dB after WSS every 5 spans	Met
OSNR performance	Above 16.5 dB (0.1 nm) required	ASE-limited OSNR 26.0 dB at 800 km (hybrid); equivalent total OSNR 18.6 dB including NLI and transceiver noise	Met
BER (pre-FEC) / Q-factor	BER ≤ 2×10^{-2} (Q ≥ 6.25 dB), 1.5 dB margin	BER 5.3×10^{-3}, Q = 8.16 dB, SNR margin 2.1 dB at 0 dBm per channel; reach with 1.5 dB margin 1,120 km	Met
DSP: chromatic dispersion	At least 13,600 ps/nm	Static FDE, about 393 equivalent taps (scalable to 981 taps for 2,000 km)	Met

Requirement / performance parameter	**Specified / target value**	**Achieved / evidence value**	**Status**
DSP: PMD / polarization	DGD of at least 8.5 ps	2×2 MIMO CMA → LMS, 21 T/2-spaced taps, covers ± 167 ps	Met
DSP: nonlinear tolerance	Controlled nonlinear penalty	Operating at near-optimum launch power (window −3.8 to +0.9 dBm within 0.5 dB); DBP optional (+0.2 dB)	Met (DBP not needed)
Standards and feasibility	ITU-T G.698 and IEEE 802.3ck compliance	50 GHz G.694.1 grid; G.698.2-style link parameters; 802.3ck / CEI-112G host interface; 400ZR-class coherent architecture (formal conformance to be tested)	Met (conformance testing pending)
Power consumption (extra)	Near 40 pJ/bit module level	About 40 pJ/bit (transceiver), about 70 pJ/bit with amplifiers (estimates)	Acceptable

6.2 Final system design

Block-level architecture



Legend: Raman pump is counter-propagating; GFF = gain-flattening filter; BPS = blind phase search; FDE = frequency-domain equalizer; ICR = integrated coherent receiver.

Fig. System block diagram of the proposed 32 x 200 Gb/s DP-16QAM DWDM link (6.4 Tb/s).

Fig. 6.1 Block diagram of the 32 × 200 Gb/s DP-16QAM DWDM link: transmitter array, AWG multiplexer, hybrid Raman/EDFA spans, AWG demultiplexer, coherent receiver (ICR), ADC and DSP chain.

Explanation of the diagram. Each of the 32 transmitters frames the 200 Gb/s client signal, adds 20% SD-FEC, maps it to DP-16QAM and shapes it digitally (root-raised-cosine, roll-off 0.1). Four DACs and drivers feed the silicon DP-IQ MZM, which modulates the ITLA carrier; the on-chip polarization rotator/combiner creates the dual-polarization signal. An isolator and attenuator level the channel, the AWG multiplexes 32 channels onto 50 GHz spacing, and the booster EDFA sets 0 dBm per channel. Each 80 km span is amplified by a backward Raman pump and a gain-flattened EDFA with isolators; a WSS equalizes channel power every five spans. The last amplifier feeds the AWG demultiplexer, which separates the channels for 32 intradyne coherent receivers, whose four photodiode signals are digitized by four ADCs and processed by the DSP chain (CD equalizer, timing recovery, MIMO, frequency offset, phase recovery, FEC).

System design parameter table

Subsystem	**Parameter**	**Value**
Channel plan	Carrier frequencies / wavelengths	192.30 – 193.85 THz (1560.06 – 1547.59 nm), 193.1 THz + n × 50 GHz, n = −16 … +15
	Grid spacing / symbol rate / format	50 GHz / 30 GBd / DP-16QAM, RRC roll-off 0.1, 240 Gb/s line rate
Transmitter laser	Output power, linewidth, RIN	+16 dBm, ≤ 100 kHz, ≤ −145 dB/Hz
Modulator (Si DP-IQ MZM)	Vπ, length, bandwidth, drive	Vπ ≈ 5 V (VπL = 2 V·cm, L = 4 mm), ≥ 30 GHz (3 dB), 3 Vpp, null-bias extinction ≥ 30 dB, loss about 19 dB total
DAC / ADC	Sampling rate, resolution	60 GSa/s (2 samples/symbol), 8-bit, effective resolution ≥ 5.5 bits
Mux / demux	AWG 32 ch	Loss 4.5 dB, adjacent-channel crosstalk ≤ −25 dB, flat-top 3 dB passband ≥ 35 GHz
Booster EDFA	Output / noise figure	15.05 dBm total (0 dBm per channel), NF 5.5 dB
Span amplifier (hybrid)	Raman gain, EDFA gain, NF	8 dB on-off (backward, 1.0 – 1.2 W, 1425 – 1485 nm) + 12 dB EDFA; NF_eff 2.0 dB (design); EDFA NF 5.5 dB
Gain control	GFF, WSS	GFF ripple < 0.7 dB p-p; WSS equalizer every 5 spans, ± 0.5 dB
Fibre (SMF-28)	Length, α, D, S	80 km × 10 spans, 0.2 dB/km, 17 ps/nm/km, about 0.058 ps/nm²/km
	γ, A_eff, PMD	1.3 $W^{-1}km^{-1}$, about 80 μm², 0.1 ps/√km
Coherent receiver (ICR)	LO linewidth, bandwidth, input	≤ 100 kHz, ≥ 25 GHz, −15 to 0 dBm per channel (about −7 dBm typical), Ge photodiode about 0.8 A/W, CMRR ≥ 20 dB
DSP	CD equalizer	Frequency-domain, about 393 equivalent taps for 800 km, FFT block 1,024 – 2,048

Subsystem	Parameter	Value
	MIMO equalizer	2×2 butterfly, 21 T/2-spaced taps, CMA/RDE then decision-directed LMS
	Carrier recovery	4th-power FFT frequency estimator (± 2.5 GHz), BPS with 32 test phases, window 10 – 20 symbols
FEC	Overhead / threshold	20% SD-FEC, pre-FEC BER ≤ 2×10^{-2}, post-FEC BER < 10^{-15}

6.3 Simulation results and validation

Method. The results below were generated with a numerical model: a Gaussian-noise (GN) model for SPM/XPM interference in a 32 × 50 GHz system with ASE noise from Eq. (6), a transceiver SNR of 17 dB, Monte-Carlo generation of 16-QAM constellations with additive Gaussian noise at the computed SNR, and a root-raised-cosine (roll-off 0.1) 4-level eye diagram of the received in-phase rail. The same system should be reproduced in OptiSystem or OptSim with the settings in Table 6.1; the screenshots from that run can replace or accompany these figures.

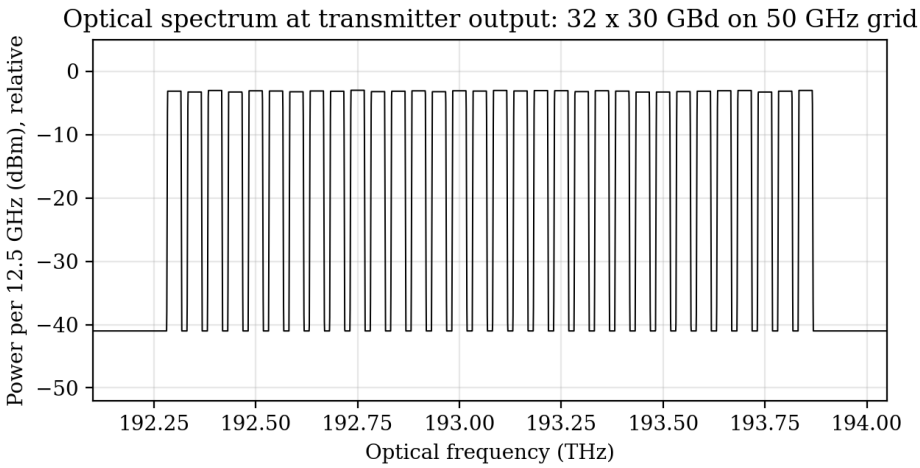


Fig. 6.2 Optical spectrum at the transmitter output: 32 channels at 30 GBd on the 50 GHz grid (relative power; modelled).

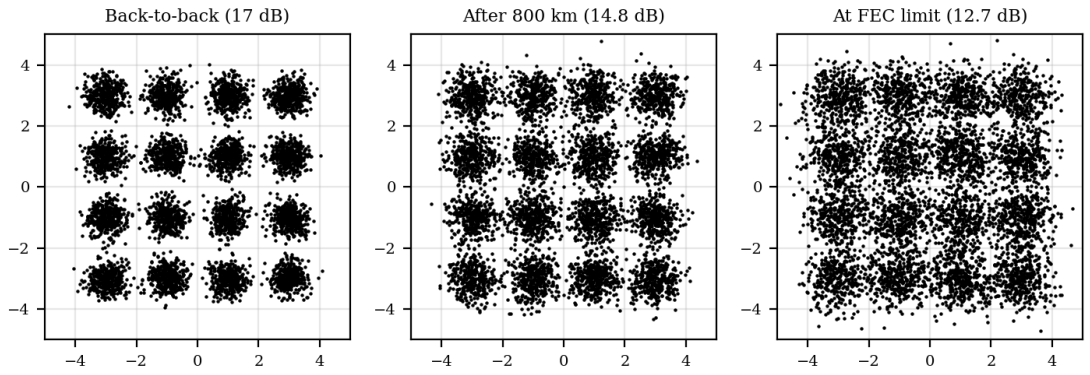


Fig. 6.3 Received DP-16QAM constellation (one polarization): back-to-back (SNR 17 dB), after 800 km (SNR 14.8 dB) and at the FEC limit (SNR 12.7 dB).

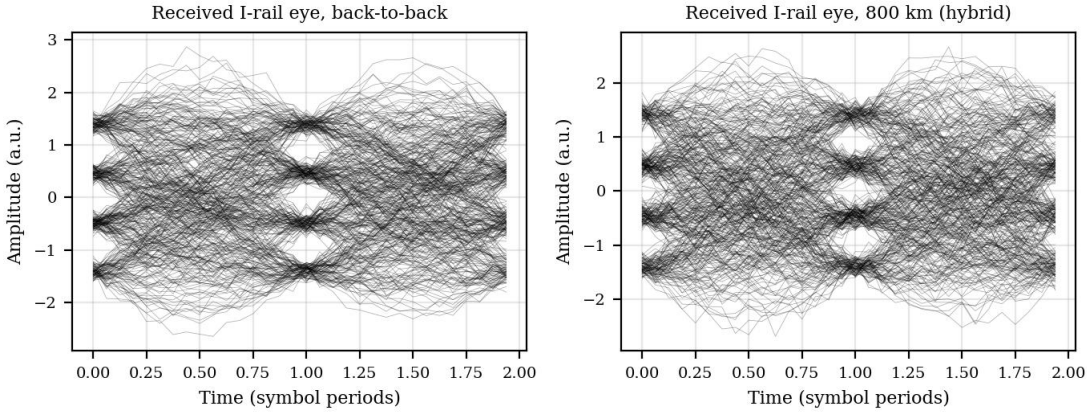


Fig. 6.4 Received eye diagram of the 4-level in-phase rail after matched filtering: back-to-back and after 800 km with the hybrid chain.

Fig. 3.3 (Q versus launch power), Fig. 3.4 (SNR versus distance), Fig. 3.2 (OSNR versus spans), Fig. 2.1 (BER versus OSNR) and Fig. 4.1 (effect of DBP) complete the validation set. The Q-versus-launch-power curves have the expected shape: Q rises with power while ASE dominates, peaks at −1.2 dBm (hybrid) or −0.1 dBm (EDFA only), and falls at higher power as nonlinear interference dominates (3 dB higher power in the nonlinear regime raises NLI by 9 dB).

Link stage	**OptiSystem / OptSim block (names vary with version)**	**Settings to enter**
Transmitter	CW laser array (32 channels); PRBS or QAM sequence generator; DP-IQ Mach-Zehnder modulator or DP-16QAM transmitter component	193.1 THz + n × 50 GHz, linewidth 100 kHz, 30 GBd, RRC roll-off 0.1, modulator Vπ 5 V, bias at null
Multiplexer	WDM multiplexer (Gaussian or AWG model)	Insertion loss 4.5 dB, bandwidth 35 GHz

Link stage	OptiSystem / OptSim block (names vary with version)	Settings to enter
Booster / span amplifiers	EDFA (black-box, NF 5.5 dB, gain 8 dB / 12 dB); Raman amplifier (backward pumping)	Output 15.05 dBm total; Raman 8 dB on-off, NF_eff 2 dB; GFF; isolators
Fibre	Optical fibre (nonlinear, with PMD)	80 km × 10 loops, 0.2 dB/km, D 17, S 0.058, γ 1.3, A_eff 80 μm², PMD 0.1 ps/√km, split-step
Demultiplexer	WDM demultiplexer	Same as multiplexer
Receiver	Coherent receiver (90° hybrid, balanced photodiodes, LO)	LO linewidth 100 kHz, responsivity 0.8 A/W, bandwidth 25 GHz; ADC 60 GSa/s, 6 bits
DSP and analysers	Coherent DSP (CD comp, MIMO, FOE, CPE) or MATLAB co-simulation; BER analyser, constellation visualiser, optical spectrum analyser, eye diagram	FDE about 393 taps, MIMO 21 taps, BPS 32 phases; sweep launch power −8 to +8 dBm to plot Q versus power

Table 6.1 Settings for reproducing the simulation in OptiSystem / OptSim / MATLAB.

FINAL ENGINEERING DECISION

Modulation format efficiency and OSNR sensitivity. DP-16QAM needs only 30 GBd and 50 GHz for 200 Gb/s, giving 4 b/s/Hz and 6.4 Tb/s in 1.6 THz of the C-band. Its required OSNR is 16.5 dB (SNR 12.7 dB) against 26.0 dB ASE-limited OSNR at 800 km and an overall SNR of 14.8 dB. The format is 3.5 times lower in baud and 2.7 times higher in efficiency than PAM4/DQPSK, and 4 dB more sensitive than DP-64QAM.

- **Photonic modulator integration, bandwidth and drive voltage.** The silicon DP-IQ MZM ($V\pi \approx 5$ V, 30 – 45 GHz) exceeds the 21 GHz needed at 30 GBd, runs from a 3 Vpp CMOS/SiGe driver, and integrates monolithically with MMI couplers, AWG and Ge photodiodes in a die of about 5 × 10 mm. The micro-ring modulator was rejected for thermal sensitivity and nonlinearity; TFLN is reserved for later upgrades.
- **Amplification gain profile, noise accumulation and power budget.** 20 dB span gain (8 dB Raman + 12 dB EDFA) balances the 20 dB span loss, with 32 mW total output within the EDFA

saturation power. The hybrid chain reduces the effective NF from 5.5 to 2.0 dB (+3.5 dB OSNR, 2.2 times as many spans), GFF plus WSS holds channel power spread within 0.5 dB, and the hybrid design is the only option that keeps the 1.5 dB margin at 800 km (EDFA-only leaves 1.2 dB).

- **DSP effectiveness against linear and nonlinear impairments.** CD of 13,600 ps/nm (108 symbols of ISI) is removed by a 393-tap-equivalent FDE, DGD of up to 8.5 ps by the 21-tap MIMO equalizer, laser phase noise and frequency offset by BPS and FOE. The nonlinear penalty is controlled by launching near the optimum power (window of 4.7 dB for 0.5 dB penalty); DBP offers only 0.2 dB and is not justified.
- **Total system power consumption, scalability and cost-effectiveness.** About 40 pJ/bit per module and about 70 pJ/bit with the 10 amplifier sites (estimates). The silicon photonic platform and 400ZR-class technology give a low unit cost; the C-band supports up to 88 channels (17.6 Tb/s), with 400G (60 GBd) and L-band upgrades available on the same fibre.

Conditions of deployment. (1) Use the hybrid Raman + EDFA configuration (EDFA-only is limited to about 640 km at 1.5 dB margin). (2) Implement WSS equalization and automatic gain control, pump-safety shutdown and automatic bias control. (3) Specify component PDL at 0.2 dB or less and transceiver SNR at or above 17 dB. (4) Verify the design by OptiSystem/OptSim simulation and conformance testing with the module vendor before deployment. **Residual risks** are Raman pump safety, PDL accumulation, converter resolution and the fact that 200 Gb/s DP-16QAM at 30 GBd is not itself a published IEEE/OIF application code. If the 1.5 dB margin cannot be demonstrated in the lab, the fall-back redesign is to reduce the span length to 70 km or to reduce reach to 8 spans; the next step up would be a 60 GBd DP-QPSK format on a 75 GHz grid (13.1 dB required OSNR) at the cost of spectral efficiency (2.67 b/s/Hz).

Submission-requirement checklist

Required report content	**Where it is covered**
1. Literature review and technology benchmarking	Stage 1, Section 1.1 and Sections 1.4 – 1.5; benchmarking tables in Stages 2 – 4
2. Modulation format and high-speed modulator trade-off calculations	Stage 2, Sections 2.1 – 2.6 (baud rate, SE, OSNR, $V\pi$, bandwidth)
3. Amplifier noise figure and span power calculations	Stage 3, Sections 3.2 – 3.5

Required report content	Where it is covered
4. Impairment mitigation and coherent DSP chain analysis	Stage 4, Sections 4.1 – 4.6
5. Simulation results, constellations, eye diagrams	Section 6.3, Figs. 3.2 – 3.4, 4.1, 6.2 – 6.4 and Table 6.1
6. Complete labelled system block diagram and parameter tables	Section 6.2, Fig. 6.1 and the parameter table
7. Verification table, final deployment decision with references	Section 6.1, Final Engineering Decision, References

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[3] IEEE Std 802.3ct-2021, Physical layers and management parameters for 100 Gb/s operation over DWDM systems.

[4] IEEE Std 802.3ck-2022, Electrical interfaces for 100 Gb/s, 200 Gb/s and 400 Gb/s per lane.

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